

1. Practice Azure_VM_2 (Linux Web Server VM)

Step 1: Create the Virtual Machine

1. In the Azure Portal, search for Virtual machines and click Create > Azure virtual machine.
2. Under Basics:
 - Subscription: Select your active subscription.
 - Resource Group: Select the existing Resource Group allocated to you.
 - Virtual machine name: WebServerVM.
 - Region: Central India.
 - Availability options: No infrastructure redundancy required.
 - Image: Ubuntu Server 20.04 LTS.
 - Size: Standard_B1s.
 - Authentication type: Select SSH public key.
 - Username: Enter your admin username (e.g., azureuser).
 - SSH public key source: Select Generate new key pair.
 - Inbound port rules: Allow HTTP (80), HTTPS (443), and SSH (22).
3. Click Review + create, then click Create and download the .pem key file.

Step 2: Configure Azure Backup

1. Navigate to the created WebServerVM resource.
2. In the left panel under Operations, click Backup.
3. Select or create a default Recovery Services Vault in Central India.
4. Choose the default backup policy and click Enable backup.

Step 3: Install Web Server (Nginx or Apache) via SSH

1. Copy the Public IP address of WebServerVM.
2. Connect to the VM via SSH from your local terminal:

Bash

```
chmod 400 /path/to/key.pem
```

```
ssh -i /path/to/key.pem azureuser@<VM_PUBLIC_IP>
```

3. Install and start Nginx:

Bash

```
sudo apt update && sudo apt install -y nginx
```

```
sudo systemctl enable --now nginx
```

Step 4: Enable Azure Security Center / Defender

1. Search for Microsoft Defender for Cloud (Azure Security Center) in the portal.
2. Go to Environment settings in the left menu.
3. Select your subscription and enable Defender for Servers (Standard Tier).
4. Click Save.

Step 5: Test Accessibility & Performance

1. Open a browser and enter `http://<VM_PUBLIC_IP>` to confirm the welcome page loads.
2. Validate response time via terminal:

Bash

```
curl -o /dev/null -s -w 'Total Time: %{time_total}s\n' http://<VM_PUBLIC_IP>
```

2. Practice Azure_VM_3 (Windows IIS Web Server VM)

Step 1: Create the Virtual Machine

1. Go to Virtual Machines > Create > Azure virtual machine.
2. Under Basics:
 - Subscription: Select your active subscription.
 - Resource Group: Select your pre-existing Resource Group.
 - Virtual machine name: winvm.
 - Region: East US.
 - Availability options: No infrastructure redundancy required.
 - Image: Windows Server 2022 Datacenter Azure Edition.
 - Size: Standard_DS1_v2.
 - Authentication type: Select Password.
 - Username / Password: Enter and confirm administrator credentials.
 - Inbound port rules: Allow RDP (3389) and HTTP (80).
3. Click Review + create and then Create.

Step 2: Install IIS Web Server

1. Once deployed, click Connect > RDP and download the RDP file.
2. Open RDP, enter your administrator username/password, and log into the VM.
3. Inside the Windows VM, open Server Manager.
4. Click Add roles and features.

5. Proceed through the wizard until Server Roles, select Web Server (IIS), add required features, and complete installation.

Step 3: Validate Web Server

1. Copy the public IP address of winvm.
2. On your local machine browser, navigate to http://<WINVM_PUBLIC_IP>.
3. Verify that the default IIS Welcome Page displays.

3. SESSION-2 COD-1 (Azure Storage Table Account)

1. Search for Storage accounts and click Create.
2. Under Basics:
 - Resource Group: Select your assigned Resource Group.
 - Storage account name: Set the name to match your Resource Group Name (lowercase, alphanumeric only).
 - Region: North Central US.
 - Primary service: Azure Blob Storage or Azure Data Lake Storage / Azure Files.
 - Performance: Standard.
 - Redundancy: Geo-redundant storage (GRS).
3. Under Advanced:
 - Access tier: Select Hot.
4. Click Review + create and then Create.
5. After deployment, navigate to the storage account resource.
6. In the left panel, scroll down to Data storage > Tables.
7. Click + Table, name it table1, and click OK.

4. SESSION-2 COD-2 (Azure Blob Storage Account)

1. Go to Storage accounts > Create.
2. Under Basics:
 - Resource Group: Select your assigned Resource Group.
 - Storage account name: Set the name to match your Resource Group Name (lowercase, alphanumeric).
 - Region: East US.
 - Performance: Standard.
 - Redundancy: Geo-redundant storage (GRS).
3. Click Review + create and then Create.

4. Go to the created Storage account and select Containers under Data storage.
5. Click + Container:
 - Name: media.
 - Anonymous access level: Select Blob (anonymous read access for blobs only).
 - Click Create.
6. Click into the media container and select Upload.
7. Browse and upload three local images (image1.jpg, image2.jpg, image3.jpg).

5. AZURE SQL Q-2 (Azure SQL Database Creation)

1. Search for SQL databases in the top bar and click Create.
2. Under Basics:
 - Database name: products.
 - Server: Click Create new:
 - Server name: Enter your Resource Group Name (lowercase, e.g., myrgname).
 - Location: West US.
 - Set Authentication to SQL authentication and enter admin credentials. Click OK.
 - Compute + storage: Click Configure database:
 - Select General Purpose.
 - Set Max size to 2 GB.
 - Click Apply.
 - Backup storage redundancy: Select Locally-redundant backup storage.
3. Under Additional settings:
 - Collation: SQL_Latin1_General_CP1_CI_AS.
4. Click Review + create and then Create.

6. AZURE SQL Q-3 (Primary & Secondary Servers with Dynamic Data Masking)

Step 1: Create Primary Server & Database (headserverdb)

1. Go to SQL databases > Create.
2. Database name: headserverdb.
3. Server: Click Create new:
 - Server name: <Your Group Name Resource>hserver.
 - Location: West US.

- Set admin login and password, click OK.
- 4. Compute + storage:
 - Select Serverless - General Purpose.
 - Set Data Max Size: 1 GB.
- 5. Backup storage redundancy: Locally-redundant backup storage.
- 6. Click Review + create and then Create.

Step 2: Create Secondary Server & Database (branchserverdb)

1. Go to SQL databases > Create.
2. Database name: branchserverdb.
3. Server: Click Create new:
 - Server name: <Your Group Name Resource>bserver.
 - Location: West US.
 - Set admin login and password, click OK.
4. Compute + storage: Select Serverless - General Purpose, Max size 1 GB.
5. Backup storage redundancy: Locally-redundant backup storage.
6. Click Review + create and then Create.

Step 3: Configure Networking Rules for Primary Server

1. Go to the primary server <Your Group Name Resource>hserver.
2. In the left panel, select Networking.
3. Check the box Allow Azure services and resources to access this server.
4. Click Add your client IPv4 address under Firewall rules.
5. Click Save.

Step 4: Create Table & Enable Dynamic Data Masking

1. Go to SQL databases > headserverdb.
2. Select Query editor (preview) from the left menu and log in.
3. Run the following query to create the schema and table:

SQL

```
CREATE SCHEMA Service;
```

```
GO
```

```
CREATE TABLE Service.customerdata(
```

```
ID INT MASKED WITH (FUNCTION = 'default()'),
CustomerNo INT MASKED WITH (FUNCTION = 'default()')
);
GO

INSERT INTO Service.customerdata (ID, CustomerNo)
VALUES (123, 12345678), (234, 543216789), (456, 987654321);
GO

```[cite: 5]
```